

Answer ALL questions. ALL working should be shown.

1. How do you describe the position of a rigid body?

[1 mark]

2. Show that the rotation matrix ${}^A_B R = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is orthonormal.

[4 marks]

3. Frame $\{B\}$ is obtained by rotating $\{A\}$ along the z -axis by 45 deg. In frame $\{B\}$, the point P has a coordinate of $[2 \ 3 \ 1]^T$. What is the coordinate of point P in frame $\{A\}$? Keep your answer in fraction instead of decimal points.

[2.5 marks]

4. Given ${}^A P = {}^A_B R \cdot {}^B P + {}^A P_{BORG}$, write the equation using homogeneous transform.

[0.75 marks]

5. Given ${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{BORG} \\ 0 & 1 \end{bmatrix}$, derive the inverse of ${}^A_B T$. Note: you will need to show the derivation, not just give the final answer from your memory.

[2.5 marks]

6. Given ${}^A_B R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$. Calculate α , β and γ .

Specify also the condition when some answers cannot be obtained. (Note: $c\alpha$ is the shorthand notation for $\cos \alpha$, $s\alpha$ is the shorthand notation for $\sin \alpha$. Similar abbreviation for β and γ .)

[3 marks]

7. What is the advantage of using Euler parameters to represent orientation?

[0.5 marks]

8. How many combinations are there for Fixed Angles representation of orientation? What are they?

[1 mark]

9. How many parameters are there in DH Parameters? And what are they (full names and symbols)?

[0.75 marks]

10. In your own words, explain the general idea of solving the forward kinematics problem using systematic method of DH parameters.

[3 marks]

11. Given the robot as shown in Fig 11.1, derive the DH parameters and summarize the results in a DH Table.

Note:

- If there is any ambiguity in up-down direction, choose up.
- If there is any ambiguity in left-right direction, choose right.
- If there is any ambiguity in into the plane / out of the plane, choose out of the plane.
- If there is any ambiguity in the diagonal direction, choose left + up.

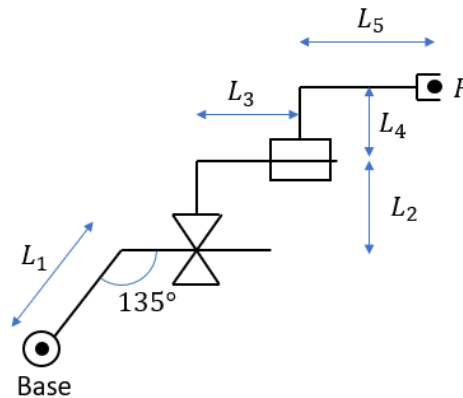


Fig. 11.1 Schematics of an RPR Robot

[7 marks]

12. A DH table is as follows (Table 12.1):

Table 12.1 DH Table

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	3	30 degrees	0	θ_1
2	2	0 degrees	d_2	30 degrees
3	1	60 degrees	1	θ_3

The transformation matrix from frame $i - 1$ to frame i is given by:

$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Calculate the transformation matrices 0_1T , 1_2T , 2_3T and 0_3T . (Note: $c\theta$ is the shorthand notation for $\cos \theta$, $s\theta$ is the shorthand notation for $\sin \theta$. Similar abbreviation for α .)

[23 marks]

13. What are the two fundamental questions which need to be answered when solving inverse kinematics problems?

[0.5 marks]

14. When would an inverse kinematics problem have NO solution?

[0.5 marks]

15. Sketch the workspace of the robot shown in Fig 15.1. Assume that the prismatic joint has a range of 0 m and 1 m, and that the revolute joints have no limits.

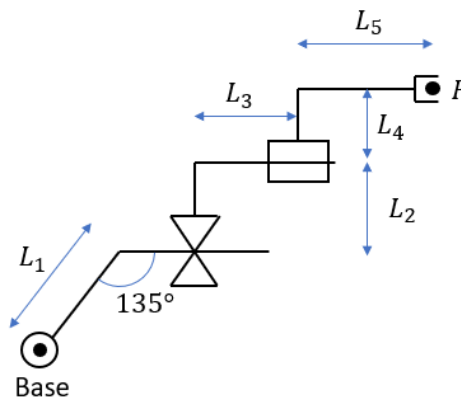


Fig 15.1 Schematics of an RPR Robot

[3 marks]

16. Given a robot as shown in Fig 16.1, derive the inverse kinematics of the robot if the end-effector is required to be at ${}^0P = [p_x \ p_y \ p_z]^T$ using the geometric approach.

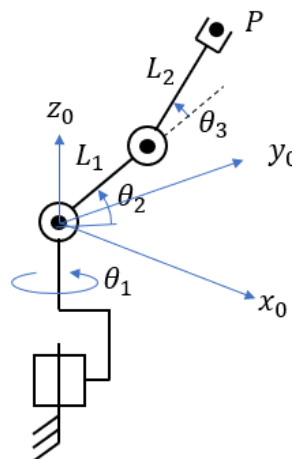


Fig 16.1 Schematics of an RRR Robot

[6 marks]

17. What does kinematic singularity mean? What is the mathematical condition for this to happen?

[1 mark]

18. The end-effector has the following position vector with respect to frame $\{0\}$:

$${}^0P = \begin{bmatrix} L_1 \cos \theta_1 \sin(\theta_1 + \theta_2) + L_2 \cos \theta_2 \\ d_3 \\ L_1 \cos \theta_1 \cos(\theta_1 + \theta_2) + d_3 \sin \theta_2 \end{bmatrix}$$

where L_i 's are constants, while θ_i 's and d_i 's are variables. Calculate the linear Jacobian matrix of the end-effector with respect to frame $\{0\}$. Does the end-effector have any singularities? If yes, please specify when singularities would happen.

[13 marks]

19. What are the disadvantages of Cartesian space schemes in trajectory planning?

[0.75 marks]

20. Explain the procedure to calculate the parameters a_0, a_1, a_2, a_3 for a cubic polynomial $u(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$, where starting position is u_0 and final position is u_f . The time to reach final position is t_f .

[5.5 marks]

21. Sketch the position, velocity and acceleration profiles for a quintic polynomial. Assume u_0 is less than u_f .

[0.75 marks]

22. Explain the general idea of calculating the dynamic equation of robots using the Newton Euler Formula.

[2.5 marks]

23. The dynamic equation for a 2-link robot is as follows:

$$\begin{aligned} \tau_1 &= m_2 L_2^2 (\ddot{\theta}_1 + \ddot{\theta}_2) + m_2 L_1 L_2 c_2 (2\ddot{\theta}_1 + \ddot{\theta}_2) + (m_1 + m_2) L_1^2 \ddot{\theta}_1^2 - m_2 L_1 L_2 s_2 \dot{\theta}_2 \\ &\quad - 2m_2 L_1 L_2 s_2 \dot{\theta}_1 \dot{\theta}_2 + m_2 g L_2 c_{12} + (m_1 + m_2) g L_1 c_1 \\ \tau_2 &= m_2 L_1 L_2 s_2 \dot{\theta}_1^2 + m_2 L_1 L_2 s_2 \ddot{\theta}_1 + m_2 g L_2 c_{12} + m_2 L_2^2 (\ddot{\theta}_1 + \ddot{\theta}_2) \end{aligned}$$

Are there any errors in the equation? If yes, please list them down. (Note: c_i is the shorthand notation for $\cos \theta_i$, s_i is the shorthand notation for $\sin \theta_i$, c_{ij} is the shorthand notation for $\cos(\theta_i + \theta_j)$, s_{ij} is the shorthand notation for $\sin(\theta_i + \theta_j)$)

[3 marks]

24. What is the equation for Coulomb friction?

[0.5 marks]

25. Given a rectangular block with length l , width w and height h as shown in Fig 25.1, calculate the mass moment of inertia I_{xx} . Assume the volume is V , the density ρ is uniform and the mass is m .

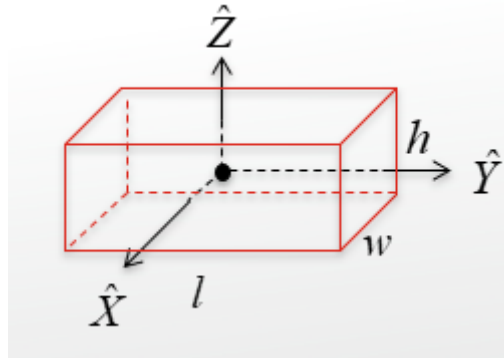


Fig. 25.1 Rectangular Block

(Hint: the formula for I_{xx} is $I_{xx} = \iiint_V (y^2 + z^2) \rho dV$.)

[5 marks]

26. Sketch a robot control block diagram, where the reference trajectory is given in Cartesian space, and the joints are controlled individually.

[1.5 marks]

27. Determine the motion of a mass-spring-damper system if the parameter values are $m = 3$, $b = 2$, $k = 4$, and the mass (initially at rest) is released from position $x = 2$.

[5 marks]

28. In what condition would a mass-spring-damper system experience an overdamped response?

[0.75 marks]

29. What happens to the robot's dynamic behaviour if we use linear control to control each joint independently?

[0.75 marks]

30. In your own words, explain why trajectory following control leads to x tracking x_d .

[1 mark]